

	Answer	Mark
2(a)(i)	The only correct answer is C <i>A is incorrect because P is a ribosome</i> <i>B is incorrect because Q is the cell wall</i> <i>D is incorrect because T is the membrane</i>	(1)

	Answer	Mark
2(a)(ii)	The only correct answer is C <i>A is incorrect because light-dependent reaction takes place on membranes and Q is the cell wall</i> <i>B is incorrect because light-dependent reaction takes place on membranes and S is the cytoplasm</i> <i>D is incorrect because light-dependent reaction takes place on membranes and U is the flagellum</i>	(1)

	Answer	Mark
2(a)(iii)	The only correct answer is B <i>A is incorrect because light-independent reaction takes place in the cytoplasm and Q is the cell wall</i> <i>C is incorrect because light-independent reaction takes place in the cytoplasm and T is the cell membrane</i> <i>D is incorrect because light-independent reaction takes place in the cytoplasm and U is a flagellum</i>	(1)

Question number	Answer	Additional guidance	Mark
2(b)	A description that includes two of the following points: • bacterial (pigments) can absorb (light) at a greater range of wavelengths (1) • bacterial (pigments) absorb more (light) at most wavelengths (1) • bacterial (pigments) absorb most (light) at 380 nm and 880 nm and plant pigments at 440 nm (1)	ACCEPT bacterial (pigments) absorb at the end of the spectrum as well but plants do not ACCEPT (overall) more absorption there are more peaks for bacterial (pigments) ACCEPT +/- 10	(2)

Question number	Answer	Additional guidance	Mark
2(c)	A description that includes the following points: - distance solvent moved (from the origin) measured (1) - distance {pigment / solute} moved from the origin (1) - equation {described / given} (1)	DO NOT ACCEPT at end of paper ACCEPT measure from {centre / leading edge} of pigment (1) $R_f = \text{distance moved by pigment} \div \text{distance moved by solvent}$	(3)